Cyclotomic Economic Unification

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Abstract

This paper develops a finite-to-continuous approximation framework for interacting cyclotomic phase states. Each state is represented by a primitive root of unity, and the aggregate and discrepancy are defined by complex addition and subtraction. Ramanujan sums provide exact Fourier coefficients for the finite phase space. A von Mises exponential tilt introduces a transparent model of relative-phase dependence, while modified Bessel functions characterize the continuous benchmark. The main result bounds the difference between the finite and continuous expectations of the squared aggregate magnitude. Prime-power orders exhibit exact Fourier cancellation outside a divisibility sublattice, yielding a computable Bessel-tail certificate. The paper also studies radial aggregate geometry, dependence regimes, finite-budget order selection, numerical reproducibility, and potential connections to economics, finance, statistics, physics, computational science, and policy analysis. These connections are presented as possible modelling interfaces rather than as empirical findings. The framework does not, by itself, establish causal, predictive, clinical, geopolitical, or welfare claims. The objective is a compact but extensible mathematical foundation on which such applications could be built and tested.

The paper ends with "The End"

Keywords: cyclotomic fields; Ramanujan sums; Fourier analysis; circular statistics; von Mises distribution; approximation theory; computational economics; aggregate dynamics.

1 Introduction

Economic systems are commonly represented through interacting prices, interest rates, exchange rates, balance-sheet positions, expectations, and institutional constraints. Some of these variables are naturally real-valued, while others can be represented by directions, phases, or normalized complex states. This paper studies a deliberately stylized model in which two states lie on the unit circle and interact through complex addition. The construction draws on cyclotomic arithmetic, harmonic analysis, and directional statistics [1, 2, 3].

The phrase *cyclotomic economic unification* is used here as a modelling programme: finite state spaces are built from primitive roots of unity, interactions are expressed algebraically, and approximation to a continuous phase model is quantified. It is not an assertion that actual economies are literally roots of unity, nor that observed interest rates must be complex. A substantive economic application would require a measurement map from data to model states and an empirical design capable of testing the mapping.

The paper focuses on four questions. First, what exact geometric identities hold for two unit-modulus states? Second, how can arithmetic properties of the cyclotomic orders control the Fourier moments of their relative phase? Third, under a circular exponential tilt, what explicit bounds relate the finite model to a continuous benchmark? Fourth, how should candidate orders be selected when direct enumeration is computationally constrained?

The contribution is methodological. Ramanujan sums convert finite arithmetic structure into exact Fourier coefficients. Prime-power orders then provide a particularly clear cancellation pattern. Bessel-series tails turn that pattern into error certificates for both the normalizing constant and the expected squared aggregate magnitude. The paper distinguishes these proved statements from conjectures about which orders are best in practice and from potential cross-disciplinary applications.

1.1 Scope and epistemic status

The mathematical claims below are derived from stated assumptions. The numerical workflow is a reproducibility protocol, not a report of a completed empirical experiment. No real economic dataset is estimated in this paper. Statements concerning finance, welfare, geopolitics, psychology, psychiatry, biology, chemistry, warfare, or cybersecurity should therefore be read as possible conceptual extensions only. A mathematical analogy is not evidence of a causal mechanism, and an approximation certificate for one statistic is not a certificate for every feature of a distribution.

2 Model definition and notation

For an integer $m \geq 2$, let the set of primitive mth roots of unity be

$$\Omega_m = \left\{ e^{2\pi i a/m} : 1 \leq a \leq m,\ \gcd(a,m) = 1 \right\}. \tag{1}$$

The cardinality of this set is Euler's totient $\varphi(m)$. Consider two phase states $X \in \Omega_m$ and $Y \in \Omega_n$. Throughout the paper, the baseline finite model samples X and Y independently and uniformly from their respective primitive-root sets. Dependence is introduced subsequently by reweighting pairs.

Define the aggregate, discrepancy, and product by

$$S = X + Y, \qquad D = X - Y, \qquad P = XY. \tag{2}$$

Since $|X| = |Y| = 1$, these variables obey exact algebraic and geometric identities. The distinction between the baseline sampling law and the reweighted law is important: the former defines Fourier coefficients, while the latter defines the dependent model whose moments are evaluated.

3 Aggregate geometry and exact identities

Write $X = e^{i\Theta}$, $Y = e^{i\Psi}$, and define the relative phase $\Delta = \Psi - \Theta$. Direct expansion gives

$$|S|^2 = (X+Y)(\overline{X}+\overline{Y}) = 2 + 2\cos\Delta, \tag{3}$$

$$|D|^2 = (X-Y)(\overline{X}-\overline{Y}) = 2 - 2\cos\Delta. \tag{4}$$

Consequently,

$$\boxed{|S|^2 + |D|^2 = 4.} \tag{5}$$

This identity holds for every pair of unit-modulus states, regardless of their orders, their dependence, or the probability law assigned to them. It is not a statistical approximation.

Let $Q = |S|^2$. Then $0 \le Q \le 4$, and the discrepancy statistic is exactly $4 - Q$. The geometric interpretation is that the aggregate and discrepancy divide a fixed squared magnitude of four. An increase in one is matched by a decrease in the other. This resembles a conservation identity in form, but it is not a physical conservation law unless the model is embedded in a physical theory with an appropriate interpretation.

The aggregate itself satisfies $|S| \le 2$. Its extreme values occur when the phases coincide or are opposite. If $Y = X$, then $S = 2X$, $D = 0$, and $|S| = 2$. If $Y = -X$, then $S = 0$, $|D| = 2$. Such cases are compatible with the primitive-root constraints only for appropriate orders and exponents; they should not be assumed for arbitrary m, n.

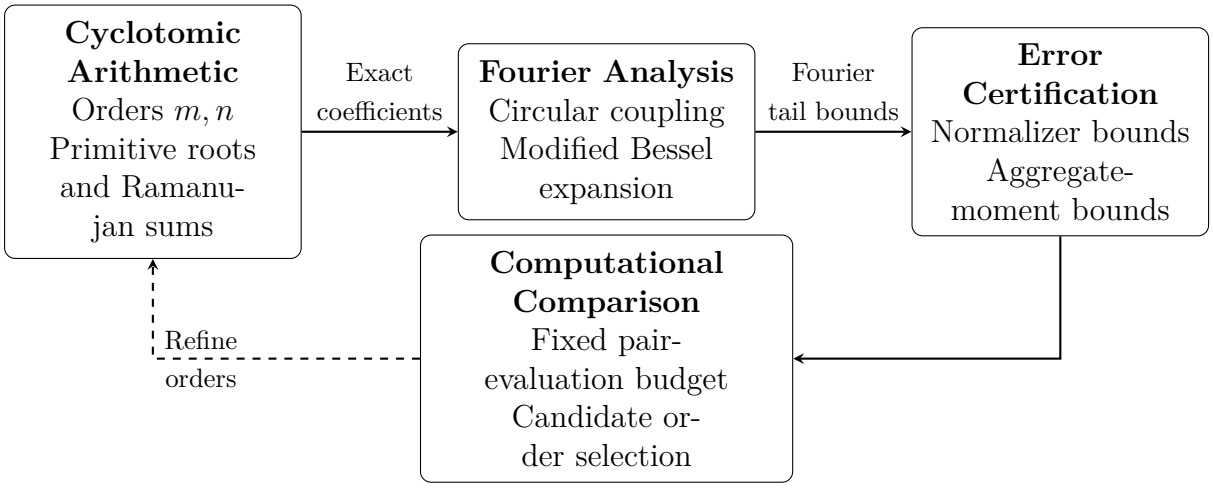


Figure 1: Analytical and computational architecture of cyclotomic economic unification. Cyclotomic arithmetic determines the finite Fourier coefficients; Fourier analysis yields approximation bounds; and computational comparisons evaluate candidate orders under a fixed enumeration budget. The feedback loop represents iterative model selection, not an additional mathematical theorem.

4 Cyclotomic arithmetic and Ramanujan sums

The Ramanujan sum of order m at integer frequency k is

$$c_m(k) = \sum_{\substack{1 \le a \le m \\ \gcd(a,m)=1}} e^{2\pi i a k/m}. \tag{6}$$

It is real-valued and can be expressed arithmetically as

$$c_m(k) = \sum_{d \mid \gcd(m,k)} d\,\mu\Big(\frac{m}{d}\Big), \tag{7}$$

where μ is the Möbius function [1, 2, 4]. The normalized Ramanujan coefficient is

$$a_m(k) = \frac{c_m(k)}{\varphi(m)}. \tag{8}$$

It follows from the triangle inequality that $|a_m(k)| \leq 1$. The coefficient records the average of the kth Fourier character over primitive roots of order m.

For a uniformly sampled primitive root $X \in \Omega_m$, one has

$$\mathbb{E}[e^{ik\Theta}] = a_m(k). \tag{9}$$

Under independent baseline sampling of X and Y, the relative phase therefore satisfies

$$\mathbb{E}_0[e^{ik\Delta}] = a_m(k)a_n(k). \tag{10}$$

This identity is the arithmetic interface at the centre of the paper. It means that the Fourier moments of the finite relative-phase distribution are known exactly from number theory. The finite-to-continuous problem is not an unspecified discretization problem: its error can be traced to explicit arithmetic coefficients.

4.1 Prime orders

For a prime p, the Ramanujan sum is $p-1$ when $p \mid k$ and -1 otherwise. Thus

$$a_p(k) = \begin{cases} 1, & p \mid k, \\ -\dfrac{1}{p-1}, & p \nmid k. \end{cases} \tag{11}$$

Most frequencies are not divisible by p, so their coefficients have magnitude $1/(p-1)$. This gives attenuation at many frequencies, but not exact cancellation.

4.2 Prime-power orders

For a prime power p^a, where $a \geq 2$, the normalized coefficient is

$$a_{p^a}(k) = \begin{cases} 0, & p^{a-1} \nmid k, \\ -\dfrac{1}{p-1}, & p^{a-1} \mid k,\ p^a \nmid k, \\ 1, & p^a \mid k. \end{cases} \tag{12}$$

The crucial feature is exact cancellation: every frequency not divisible by p^{a-1} has coefficient zero. Increasing the exponent can move the first potentially nonzero frequency to a higher location. For a pair of distinct prime powers $m = p^a$ and $n = q^b$, a product coefficient can be nonzero only when both divisibility conditions hold, equivalently when $p^{a-1}q^{b-1} \mid k$.

4.3 Squarefree composite orders

Let $m = p_1 p_2 \cdots p_r$ be squarefree, with distinct prime factors. Ramanujan sums are multiplicative in coprime orders, so

$$a_m(k) = \prod_{j=1}^{r} a_{p_j}(k). \tag{13}$$

A frequency divisible by some of the prime factors receives a coefficient determined by the remaining factors. When k is divisible by none of them, the coefficient is

$$a_m(k) = (-1)^r \prod_{j=1}^{r} \frac{1}{p_j - 1}. \tag{14}$$

This can be small, but it is generally nonzero. Thus squarefree composites and prime powers have distinct approximation mechanisms: multiplicative attenuation versus exact divisibility cancellation.

Table 1: Arithmetic families and their Fourier structures.

Order family	Normalized coefficient	Approximation mechanism
Prime p	1 if $p \mid k$; otherwise $-1/(p-1)$	Attenuation at many frequencies
Prime power p^a	Zero unless $p^{a-1} \mid k$	Exact cancellation
Squarefree $\prod_j p_j$	Product of prime coefficients	Multiplicative attenuation

5 The continuous phase limit

The continuous benchmark arises when the primitive-root phases are replaced by uniform circular phases. If Θ and Ψ are independent and uniform on the circle, then Δ is uniform modulo 2π. In this case, the aggregate magnitude is

$$R = |X + Y| = 2\left|\cos\left(\frac{\Delta}{2}\right)\right|. \tag{15}$$

For $0 \le r \le 2$, its cumulative distribution function and density are

$$F_R(r) = \frac{2}{\pi} \arcsin\left(\frac{r}{2}\right), \tag{16}$$

$$f_R(r) = \frac{2}{\pi\sqrt{4 - r^2}}, \qquad 0 < r < 2. \tag{17}$$

The radial density is not the same as the planar density of the complex aggregate. The latter, with respect to area measure in the disk, is

$$f_S(z) = \frac{1}{\pi^2 |z| \sqrt{4 - |z|^2}}, \qquad 0 < |z| < 2. \tag{18}$$

Both formulas have integrable singularities. The singularities do not imply a point mass at the origin or boundary in the continuous independent-phase model.

The distinction between radial and planar distributions is important for applications. A statistic based on the aggregate magnitude uses the radial law, while a statistic sensitive to the direction of the aggregate requires the full planar distribution. The approximation theorem developed below controls a moment of the radial distribution, not the entire planar law.

6 Von Mises coupling

To model relative-phase dependence, introduce the positive weight

$$w_{\kappa,\mu}(\Delta) = e^{\kappa \cos(\Delta-\mu)}, \tag{19}$$

where $\kappa \geq 0$ controls concentration and $\mu \in [-\pi, \pi]$ is the preferred relative phase. The corresponding continuous density is

$$f_\Delta(\delta) = \frac{e^{\kappa \cos(\delta-\mu)}}{2\pi I_0(\kappa)}, \qquad -\pi \leq \delta \leq \pi, \tag{20}$$

where I_j denotes the modified Bessel function of the first kind [3, 5].

The parameter $\kappa = 0$ recovers the uniform relative phase. Larger values concentrate probability around $\Delta = \mu$ modulo 2π. In the limiting regime of large concentration, the phase becomes increasingly concentrated near the preferred direction. These are statistical properties of the chosen model, not assertions about how real economic agents form expectations.

Let $\mathbb{E}_0$ denote expectation under independent uniform sampling from $\Omega_m \times \Omega_n$. Define the finite normalizer and numerator by

$$Z_{m,n}(\kappa, \mu) = \mathbb{E}_0[w_{\kappa,\mu}(\Delta)], \tag{21}$$

$$N_{m,n}(\kappa, \mu) = \mathbb{E}_0[Q w_{\kappa,\mu}(\Delta)]. \tag{22}$$

The finite weighted expectation of the squared aggregate magnitude is

$$M_{m,n}(\kappa, \mu) = \frac{N_{m,n}(\kappa, \mu)}{Z_{m,n}(\kappa, \mu)}. \tag{23}$$

The continuous benchmark follows from the first trigonometric moment of the von Mises distribution:

$$M_\infty(\kappa, \mu) = 2 + 2\frac{I_1(\kappa)}{I_0(\kappa)} \cos \mu. \tag{24}$$

In particular, $0 \leq M_\infty \leq 4$. At $\kappa = 0$, the benchmark equals two. As concentration grows, the benchmark approaches four when the preferred phase is zero and approaches zero when the preferred phase is π.

7 Fourier expansion and normalizer error

The Bessel expansion of the von Mises kernel is

$$e^{\kappa \cos(\Delta-\mu)} = \sum_{k \in \mathbb{Z}} H_k e^{ik\Delta}, \qquad H_k = I_{|k|}(\kappa) e^{-ik\mu}. \tag{25}$$

The continuous normalizer is $I_0(\kappa)$. In the finite model, equation (10) gives

$$Z_{m,n}(\kappa, \mu) = \sum_{k \in \mathbb{Z}} H_k a_m(k) a_n(k), \tag{26}$$

where $a_m(0) = a_n(0) = 1$. The zero-frequency term is $I_0(\kappa)$; all discrepancy from the continuous normalizer is therefore contributed by nonzero frequencies.

Define

$$E_Z(m,n;\kappa) = 2\sum_{k=1}^{\infty} I_k(\kappa)|a_m(k)a_n(k)|. \tag{27}$$

Absolute convergence holds for every fixed finite κ. Applying the triangle inequality yields

$$|Z_{m,n}(\kappa,\mu) - I_0(\kappa)| \le E_Z(m,n;\kappa). \tag{28}$$

The bound is uniform in μ, because the location parameter contributes only unit-modulus Fourier factors.

If $E_Z < I_0(\kappa)$, then the finite normalizer is bounded away from zero:

$$Z_{m,n}(\kappa,\mu) \ge I_0(\kappa) - E_Z(m,n;\kappa) > 0. \tag{29}$$

This lower bound is essential when ratios of finite numerators and denominators are compared. A numerator estimate alone does not control a normalized expectation if the denominator is not controlled.

8 Aggregate numerator error

Since $Q = 2+e^{i\Delta}+e^{-i\Delta}$, multiplication of the kernel expansion by Q gives the coefficients

$$B_k = 2H_k + H_{k-1} + H_{k+1}. \tag{30}$$

The continuous numerator is $I_0(\kappa)M_\infty(\kappa,\mu)$. Define

$$E_N(m,n;\kappa,\mu) = \sum_{\substack{k\in\mathbb{Z}\\k\neq 0}} |B_k|\,|a_m(k)a_n(k)|. \tag{31}$$

Then

$$|N_{m,n}(\kappa,\mu) - I_0(\kappa)M_\infty(\kappa,\mu)| \le E_N(m,n;\kappa,\mu). \tag{32}$$

The numerator coefficients differ from the kernel coefficients because the statistic Q shifts frequencies by zero and by one. This shift explains the adjacent Bessel terms that appear in the prime-power certificate.

9 Main theorem: Approximation of aggregate magnitude

Theorem 1 (Finite-to-continuous aggregate-magnitude bound)**.** *Suppose $E_Z(m,n;\kappa) < I_0(\kappa)$. Then*

$$|M_{m,n}(\kappa,\mu) - M_\infty(\kappa,\mu)| \le \frac{E_N + |M_\infty|E_Z}{I_0(\kappa) - E_Z} \le \frac{E_N + 4E_Z}{I_0(\kappa) - E_Z}, \tag{33}$$

where the arguments of E_N, E_Z, and M_∞ are suppressed in the displayed expression.

Proof. *By equation (29), the denominator in equation (23) is positive. We write*

$$|M_{m,n} - M_\infty| = \frac{|N_{m,n} - M_\infty Z_{m,n}|}{Z_{m,n}} \tag{34}$$

$$\le \frac{|N_{m,n} - I_0 M_\infty| + |M_\infty||I_0 - Z_{m,n}|}{I_0 - E_Z}. \tag{35}$$

Applying equations (28) and (32) proves the first inequality. Since $M_\infty \in [0,4]$, the second inequality follows. □

The theorem provides an error bound for one particular statistic, namely the expectation of the squared aggregate magnitude. It does not imply that the full distributions are close in total variation, Wasserstein distance, or any other probability metric. It also does not directly bound a tail probability, quantile, variance, or conditional expectation. Those quantities require their own test functions and corresponding Fourier coefficients.

10 Prime-power cancellation and a uniform certificate

Let $m = p^a$ and $n = q^b$, where $p \neq q$ are primes and $a, b \geq 2$. Define

$$L = p^{a-1} q^{b-1}. \tag{36}$$

By equation (12), the product $a_m(k)a_n(k)$ vanishes unless $L \mid k$. It follows that

$$E_Z(m,n;\kappa) \leq 2 \sum_{\ell=1}^{\infty} I_{\ell L}(\kappa), \tag{37}$$

$$E_N(m,n;\kappa,\mu) \leq 2 \sum_{\ell=1}^{\infty} \left[2I_{\ell L}(\kappa) + I_{\ell L-1}(\kappa) + I_{\ell L+1}(\kappa) \right]. \tag{38}$$

These inequalities use $|a_m(k)a_n(k)| \leq 1$ and are therefore sufficient but potentially conservative. Sharper bounds can retain the exact coefficient magnitudes at the surviving frequencies.

Fix a concentration range $0 \leq \kappa \leq K$, with $K > 0$, and define

$$\overline{E}_Z(L,K) = 2 \sum_{\ell=1}^{\infty} I_{\ell L}(K), \tag{39}$$

$$\overline{E}_N(L,K) = 2 \sum_{\ell=1}^{\infty} \left[2I_{\ell L}(K) + I_{\ell L-1}(K) + I_{\ell L+1}(K) \right]. \tag{40}$$

For nonnegative arguments, the integer-order modified Bessel functions used here are nondecreasing in the argument. Hence these bounds hold uniformly for all $\kappa \in [0,K]$. If $\overline{E}_Z(L,K) < 1$, then $I_0(\kappa) \geq 1$ and Theorem 1 yields

$$\sup_{\substack{0 \leq \kappa \leq K \\ -\pi \leq \mu \leq \pi}} |M_{p^a,q^b}(\kappa,\mu) - M_\infty(\kappa,\mu)| \leq \frac{\overline{E}_N(L,K) + 4\overline{E}_Z(L,K)}{1 - \overline{E}_Z(L,K)}. \tag{41}$$

Theorem 2 (Uniform prime-power certificate)**.** *For distinct prime-power orders satisfying the assumptions above, equation (41) is a sufficient uniform error certificate over the compact parameter set $[0,K] \times [-\pi,\pi]$.*

Proof. *Divisibility cancellation restricts the nonzero terms in the Fourier sums to frequencies $k = \ell L$. Bounding the surviving normalized coefficients by one yields equations (37) and (38). Monotonicity of the Bessel terms on the specified range replaces κ by K. Finally, $I_0(\kappa) \geq 1$ and the assumption $\overline{E}_Z < 1$ give a denominator lower bound of at least $1 - \overline{E}_Z$. Substitution into Theorem 1 completes the proof.* □

The result explains why prime powers are analytically attractive: they create a gap in the set of frequencies that can contribute to the error. For fixed K, Bessel coefficients at sufficiently high orders become small, so the tail can be suppressed by increasing the first allowed frequency. If K grows at the same time as the orders, however, the Bessel mass can shift toward higher frequencies. No universal power law relating order size and concentration is asserted here.

11 Finite-budget order selection

Direct enumeration of all pairs in $\Omega_m \times \Omega_n$ requires

$$C(m,n) = \varphi(m)\varphi(n) \tag{42}$$

pair evaluations. For prime-power orders, this becomes

$$C(p^a, q^b) = p^{a-1}(p-1)q^{b-1}(q-1). \tag{43}$$

A natural design problem is to minimize the approximation error subject to a computational budget $C_{\max}$:

$$\min_{\substack{m,n\geq 2 \\ \varphi(m)\varphi(n)\leq C_{\max}}} \sup_{\substack{0\leq\kappa\leq K \\ -\pi\leq\mu\leq\pi}} |M_{m,n}(\kappa,\mu) - M_\infty(\kappa,\mu)|. \tag{44}$$

This is a finite search when a finite candidate set or an upper bound on the orders is specified. In practice, the objective may be evaluated using exact Fourier coefficients and truncated Bessel series with certified tail tolerances.

One possible diagnostic is error per pair-evaluation cost. For a chosen error certificate $\mathcal{E}(m,n;K)$, define

$$\mathcal{J}(m,n;K) = \frac{\mathcal{E}(m,n;K)}{\varphi(m)\varphi(n)}. \tag{45}$$

This ratio is a comparison device, not a universal optimality criterion. The actual design should consider absolute error, cost, numerical stability, the target statistic, and the range of parameters of interest. A smaller ratio need not imply a smaller absolute error or faster wall-clock execution in every implementation.

11.1 Candidate selection procedure

A practical selection pipeline can proceed as follows. First, define a maximum concentration K, a target tolerance ε, and a pair-evaluation budget $C_{\max}$. Second, generate candidate orders and remove those violating the budget. Third, calculate exact normalized Ramanujan coefficients for frequencies up to a truncation point. Fourth, bound the omitted Bessel tail. Fifth, evaluate the normalizer and aggregate-magnitude certificates. Finally, rank candidates by the chosen error criterion and report both the bound and the cost.

Table 2: Suggested decision criteria for finite cyclotomic designs.

Criterion	Quantity	Purpose
Pair budget	$\varphi(m)\varphi(n)$	Bound direct enumeration cost
Normalizer error	E_Z	Certify denominator stability
Numerator error	E_N	Control weighted aggregate numerator
Moment error	Right side of equation (33)	Compare the target statistic
Robustness	Supremum over κ, μ	Avoid tuning to one parameter point

Prime powers are promising because of exact cancellation. Squarefree orders may be competitive when their multiplicative attenuation is strong. Neither family is proved universally optimal under every budget, concentration range, and loss function. The optimal family is a question for comparative analysis under declared constraints.

12 Algorithmic protocol and reproducibility

The following pseudocode gives a minimal reproducible structure for comparing candidate orders. It is deliberately expressed with a standard verbatim environment rather than a specialized algorithm package.

```
INPUT: candidate order pairs (m,n), concentration grid K_grid,
       phase grid MU_grid, pair budget C_max, tolerance tol

FOR each candidate pair (m,n):
    IF phi(m)*phi(n) > C_max:
        continue
    Compute a_m(k), a_n(k) from exact Ramanujan sums
    Select a Fourier cutoff H and certify the omitted Bessel tail
    FOR each kappa in K_grid:
        FOR each mu in MU_grid:
            Compute finite normalizer Z and numerator N
            Compute E_Z and E_N, including tail bounds
            IF I_0(kappa) - E_Z <= 0:
                mark certificate inconclusive
            ELSE:
                M_finite = N/Z
                        M_limit = 2 + 2*I1/I0*cos(mu)
                Bound = (EN + 4*EZ)/(I0-EZ)
                record error, bound, cost, and settings
RETURN candidate ranking and complete diagnostics
```

The protocol has several implementation details that should be reported. Arithmetic coefficients should be computed exactly where possible, particularly for divisibility decisions. Bessel terms should be evaluated at adequate numerical precision. Truncation should stop only when the omitted tail is bounded below a declared tolerance. A numerical estimate and a theorem-based certificate should be stored separately. If Monte Carlo

is used to validate the calculation, random seeds, sample sizes, and confidence intervals should be recorded.

A reproducible report should distinguish three objects:

1. the exact finite sum implied by the chosen orders;
2. a numerical approximation to that sum, including floating-point and truncation error;
3. a rigorous upper bound on the finite-to-continuous discrepancy.

These objects answer different questions. Agreement between a simulation and a finite sum validates an implementation but does not, by itself, prove the continuous approximation bound. Conversely, a conservative certificate may be mathematically valid even when it is not numerically tight.

13 Statistical interpretation and further metrics

The main theorem controls $\mathbb{E}[Q]$, where $Q = |S|^2$. Since $0 \le Q \le 4$, the statistic is bounded and its expectation is well-defined under every probability law on the state space. The bound can be useful as a first diagnostic of aggregate behaviour. However, two distributions can share the same expectation while differing substantially in variance, quantiles, multimodality, or tail behaviour.

For example, the variance requires a bound on the second moment $\mathbb{E}[Q^2]$. Because

$$Q^2 = (2 + 2\cos\Delta)^2 = 6 + 8\cos\Delta + 2\cos(2\Delta), \tag{46}$$

its Fourier representation involves frequencies zero, one, and two. A corresponding numerator certificate can be constructed by multiplying the Bessel expansion by this trigonometric polynomial. The same general method extends to any bounded trigonometric polynomial in the relative phase.

For a general bounded continuous function of the phase, Fourier truncation or approximation by trigonometric polynomials can provide a route to distributional comparison. To obtain a bound in a named probability metric, the metric's test-function class and regularity requirements must be addressed explicitly. The present moment certificate should not be relabelled as a total-variation or Wasserstein bound without that additional work.

13.1 Sensitivity to the concentration parameter

The concentration parameter κ changes both the weighting kernel and the relative importance of Fourier frequencies. At small κ, the Bessel expansion places most weight on low orders. As κ increases, higher-order terms become more relevant. Prime-power cancellation is therefore particularly interpretable through the relation between the first allowed frequency L and the concentration scale. The useful regime is not determined by L alone; it depends on the Bessel tail at the relevant K.

The preferred phase μ changes the complex phases of the Fourier coefficients but not their absolute values in the normalizer bound. Consequently, the normalizer certificate is uniform in μ. The numerator bound is also uniform after replacing its coefficient magnitudes by the Bessel envelope used in equation (38). Uniformity is valuable when a model must remain reliable across a range of phase alignments rather than at one calibrated point.

14 Economic and financial interpretation

A possible economic interpretation is to regard X and Y as normalized phase representations of two interacting state variables. Depending on a separately justified application, these might encode cyclical positions, normalized returns, relative market states, or latent components of a financial system. The aggregate S records their complex sum, while D records their difference. The product $P = XY$ can encode combined phase information.

This interpretation requires care. An observed nominal or real interest rate is ordinarily a real scalar and does not automatically lie on the unit circle. To use the model, an analyst must define a transformation from observed quantities to phases, explain how the transformation preserves relevant economic information, and test whether the transformed variables have predictive or explanatory value. Different transformations may induce different phase dependence and hence different conclusions.

In finance, the aggregate could be treated as a stylized exposure or state-combination statistic. The identity $|S|^2 + |D|^2 = 4$ would then describe the geometry of normalized states, not the conservation of money, wealth, or risk. It does not imply that systemic risk can be hedged away, nor does it estimate a risk premium. Financial applications would need explicit cash-flow definitions, pricing assumptions, market data, and a validation strategy.

In econometrics, the framework can motivate a finite-state approximation to a circular latent-variable model. Estimation would require an observation equation and an identified likelihood or moment condition. A credible study should compare against simple baselines, account for serial dependence and structural breaks where relevant, report parameter uncertainty, and reserve out-of-sample periods for evaluation. A low theoretical approximation error for one statistic does not establish forecasting superiority.

In welfare economics, the aggregate statistic is not a welfare function. Welfare analysis would require preferences, constraints, feasible allocations, and a normative criterion. The geometric identity could be one component of a model, but no Pareto improvement, social-welfare ranking, or distributional conclusion follows from it alone.

15 Interdisciplinary interfaces and limits

15.1 Physics and engineering

Complex phases are familiar in oscillatory systems, wave analysis, and signal processing. The sum of two unit phasors is a standard geometric operation, and the aggregate–discrepancy identity follows from elementary complex algebra. In engineering, this may be useful for testing finite phase grids, coupled oscillators, or signal representations. Any physical interpretation would nevertheless require a model-specific account of the state variables, measurement process, and dynamics. The mathematical identity is not, by itself, a physical conservation law.

15.2 Chemistry and molecular science

Some molecular and spectroscopic models use complex amplitudes or phase-sensitive representations. A cyclotomic discretization could be considered where a finite phase grid is a justified numerical approximation. It would be inappropriate to infer molecular mechanisms or chemical reaction outcomes from the present model without domain equations,

experimental observations, and a validated mapping from molecular states to phase variables.

15.3 Biology, psychology, and psychiatry

Circular statistics can describe periodic measurements such as rhythmic activity, directional behaviour, or phase differences between oscillatory signals. A von Mises model may be a candidate distribution when its assumptions fit the data. Biological and behavioural applications would require measurement validity, appropriate treatment of repeated observations, ethical oversight where applicable, and external validation. This paper proposes no clinical instrument, diagnosis, or psychiatric theory, and it does not imply that human mental states are roots of unity.

15.4 Computer science, data science, and AI/ML

The arithmetic coefficients provide structured features that could be incorporated into numerical algorithms or learning systems. A machine-learning model might learn mappings from observed covariates to concentration and preferred-phase parameters, or use Fourier coefficients as interpretable features. Such a model would need a training objective, a leakage-resistant evaluation design, calibration, and comparison with non-cyclotomic baselines. A mathematically elegant feature representation is not evidence of improved predictive performance.

For efficient computation, the main choices include direct enumeration of primitive-root pairs, Fourier summation, and hybrid methods. Direct enumeration is transparent but scales as $\varphi(m)\varphi(n)$. Fourier methods exploit the known arithmetic coefficients and may be much cheaper when only a few statistics are required. Hybrid methods can use direct enumeration for small orders to verify a Fourier implementation and use certified Fourier tails for larger orders.

15.5 Cybersecurity and numerical assurance

In cybersecurity, a formal finite-state model can be useful as a conceptual tool for exploring cyclic state spaces or checking invariants. It does not establish cryptographic security. A security claim would require a threat model, adversarial analysis, implementation review, and appropriate cryptographic assumptions. Similarly, numerical assurance should include overflow checks, stable Bessel evaluation, truncation certificates, unit tests for Ramanujan coefficients, and independent verification of selected cases by direct enumeration.

15.6 Political science, diplomacy, geopolitics, and warfare economics

The framework could be used as a toy model for interacting institutions or state-level indicators if a defensible mapping to circular variables were developed. The current mathematics does not predict alliance formation, deterrence, escalation, sanctions outcomes, military conflict, or diplomatic bargaining. Those questions involve strategic choice, incomplete information, institutions, history, material capabilities, and other variables that are absent from the present formalization. A geopolitical application would require

an explicit strategic model, observable data, counterfactual assumptions, and sensitivity analysis.

Warfare economics and security policy are high-consequence settings. A compact aggregate statistic should not be used as an operational recommendation or a substitute for domain expertise. In such applications, the appropriate role of the mathematics would be limited to a transparent, independently validated component of a larger decision-support framework.

15.7 Philosophy, religion, journalism, and public communication

The paper illustrates a general epistemic distinction: an algebraic identity, a probabilistic theorem, an empirical result, and a normative interpretation are different kinds of claims. Responsible communication should identify which kind of claim is being made and what evidence supports it. The mathematics neither requires nor establishes a religious doctrine, and it does not settle philosophical disputes about agency, institutions, or social order.

For journalism and public communication, the most important safeguard is to avoid overstating what the result means. The correct statement is that the paper derives a sufficient bound for a specified statistic under a specified finite phase model. It would be inaccurate to claim that the paper has discovered a universal law of economic unification or demonstrated an empirically validated predictor of crises.

16 Limitations and open research questions

Several limitations delimit the results. First, the unit-circle representation is a modelling assumption. Its economic meaning depends on how the phase variables are constructed. Second, the finite coupling is implemented through exponential reweighting of a baseline product measure. It is not automatically identical to a model in which both finite marginals are constrained to remain exactly uniform after reweighting. Third, the error certificates use absolute values and can be conservative. Fourth, the main theorem controls one moment, not the full distribution. Fifth, no real-world data analysis is reported.

These limitations suggest a research agenda. One direction is to compare prime, prime-power, and squarefree orders under matched pair-evaluation budgets. Another is to derive sharper certificates that retain the exact normalized coefficients at the surviving frequencies. A third is to bound additional statistics such as the variance of Q, tail probabilities, or smooth test functions of the aggregate. A fourth is to investigate convergence of the entire aggregate distribution under appropriate growth conditions on the cyclotomic orders. A fifth is to define and estimate a measurement map for a particular economic dataset, then evaluate the resulting model against credible baselines.

An especially important open question is whether prime-power orders are optimal for any clearly specified class of losses and computational budgets. Exact cancellation makes them attractive, but optimality does not follow from cancellation alone: totient cost, surviving coefficient magnitudes, concentration, and the target statistic all matter. A theorem of optimality would need to compare the candidate family against all admissible alternatives under a precise objective.

17 Conclusion

This paper develops a mathematical route from cyclotomic arithmetic to finite-to-continuous approximation guarantees. Primitive roots define the finite state spaces. Ramanujan sums give exact Fourier moments. Von Mises coupling supplies a tractable continuous benchmark. The resulting Fourier bounds control the normalizer and the expected squared aggregate magnitude. Prime-power orders are particularly useful because their normalized Ramanujan coefficients vanish outside a divisibility sublattice, leading to a uniform Bessel-tail certificate over bounded concentration and phase-location ranges.

The results establish a method, not a universal empirical law. The finite-budget optimization problem remains open in general, and the superiority of one arithmetic family depends on the chosen loss and constraints. Extensions to other statistics are possible by changing the trigonometric multiplier and deriving corresponding Fourier bounds. Applications in economics, finance, engineering, machine learning, and policy require a defensible measurement map, suitable data, identification, uncertainty analysis, and independent validation. Maintaining these distinctions is part of the mathematical result's value: the framework offers explicit assumptions, inspectable derivations, and computable certificates rather than unsupported claims.

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Glossary

Aggregate
: The complex sum $S = X + Y$ of two unit-modulus states.

Aggregate magnitude
: The nonnegative quantity $|S|$, bounded above by two.

Baseline measure
: The product distribution under which the two primitive roots are sampled independently and uniformly.

Bessel tail
The sum of modified Bessel coefficients beyond a specified frequency threshold.

Computational budget
The maximum allowed number of primitive-root pair evaluations.

Cyclotomic order
The integer m specifying the exact multiplicative order of a primitive mth root of unity.

Discrepancy
The complex difference $D = X - Y$.

Error certificate
A computable upper bound on the difference between a finite statistic and its continuous benchmark.

Fourier frequency
An integer index k specifying the character $e^{ik\Delta}$.

Modified Bessel function
The function I_k appearing in the Fourier expansion of the von Mises kernel.

Normalized Ramanujan coefficient
The ratio $a_m(k) = c_m(k)/\varphi(m)$.

Primitive root of unity
A root of unity whose multiplicative order is exactly the stated integer.

Ramanujan sum
The sum of the kth powers of all primitive roots of order m, denoted $c_m(k)$.

Squarefree order
An integer divisible by no square of a prime.

Uniform certificate
An error bound that holds throughout a declared parameter set rather than at one parameter point.

Von Mises coupling
A circular exponential tilt governed by concentration κ and preferred phase μ.

The End